

Mini Project (22UEE528P) Report on

**AI-ML BASED PREDICTIVE MAINTAINANCE AND CHARGING
OPTIMIZATION OF EV BATTERY (Li-ion)**

*Submitted in the partial fulfilment for the award of degree in
Bachelor of Engineering in the
Department of Electrical and Electronics Engineering*

Submitted by:

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Under the Guidance of

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Submitted to



Department of Electrical & Electronics Engineering

[Govt. Aided, Accredited by NBA 2003, 2006, 2018, Recognized R&D centre of VTU]

Basaveshwar Engineering College (Autonomous)

[AICTE Recognized, TEQIP-I,II&III, Permanently Affiliated to VTU Belagavi, NBA & NAAC Accredited]

Bagalkot-587103, Karnataka state, India



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Department of Electrical and Electronics Engineering

Certificate

This is to certify that Mini Project entitled “**AI-ML BASED PREDICTIVE MAINTAINANCE AND CHARGING OPTIMIZATION OF EV BATTERY (Li-ion)**” is a Bonafide work of:

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The report satisfies the academic requirements in respect of project work prescribed for V semester **Mini Project (22UEE528P)** for award of Bachelor of Engineering degree in the department of Electrical and Electronics Engineering during the academic year 2025-2026.

Hereby declare that the submitted mini project work report is not submitted elsewhere.

Prof. S. S. Kandagal
Project Guide

Prof. S. S. Kandagal
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Dr. B. R. Hiremath
Principal

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1.	
2.	
3.	

Mini Project COS

At the end of this course students should be able to,

1. Identify, formulate & analyze the engineering problems associated with electrical & electronics engineering and interdisciplinary research.
- 2 Design & implement proposed solutions for complex engineering problems to meet specified objectives by analyzing / validating the design / solutions of engineering problems using contemporary tools & resources. Identify optimization opportunities to enhance reliability and sustainability.
3. Prepare engineering documents and make effective presentation to communicate effectively and collaboratively with detailed analysis and interpretation of results to yield valid conclusions.
4. Demonstrate social, ethical cultural & engineering professional responsibilities.

PROGRAM OUTCOMES

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conductive estimations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** understand the impact of the professional engineering solutions in societal and environmental contexts, demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics, responsibilities, and norms of the engineering practice.
9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Acknowledgment

Gratitude takes three forms- “A feeling from heart, an expression in words and a giving in return”. We take this opportunity to express our feelings.

We express our sincere and humble gratitude to our project guide, **Prof. Santhoshkumar. S. Kandagal**, for his valuable guidance and encouragement given during the course of the project work.

We would like to convey our thanks to the Project Coordinator **Prof. Santhoshkumar. S. Kandagal**, and we extend our sincere requital to all the faculty and staff of the Electrical and Electronics Department.

We are thankful to **Dr. R. L. Naik** HoD, Department of Electrical and Electronics Engineering for providing the facilities in the department to do project work.

We express our sincere and humble gratitude to **Dr. B. R. Hiremath** Principal, Basaveshwar Engineering College (Autonomous), Bagalkote for providing a congenial environment in the college.

A heart full thanks to our parents and to all our family members whose patience and kind cooperation at all the stages enabled us to complete the dissertation successfully.

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ABSTRACT

Electric Vehicles (EVs) have emerged as a promising solution to address environmental pollution, fossil fuel depletion, and climate change. The advancement of lithium-ion battery technology has significantly contributed to the growth of electric mobility; however, battery safety, reliability, and degradation remain critical challenges affecting large-scale EV adoption. Conventional battery monitoring techniques and hardware-based Battery Management Systems (BMS) rely on fixed thresholds and reactive control strategies, which are often complex, costly, and unsuitable for small-scale and low-power EV applications. Moreover, inefficient charging practices and the absence of predictive capability accelerate battery aging and increase the risk of unexpected failures.

Recent research has explored Artificial Intelligence (AI) and Machine Learning (ML) techniques for battery health estimation and fault detection, demonstrating improved accuracy over traditional methods. However, many existing solutions require large datasets, complex models, and high computational resources, limiting their practical deployment in cost-sensitive systems. Additionally, limited work has focused on integrating predictive maintenance with adaptive charging control using simple and lightweight charging modules.

In this context, this paper proposes an AI/ML-based predictive maintenance and charging optimization system for EV batteries with a simplified and cost-effective design. The proposed system monitors key battery parameters such as voltage, current, and temperature using sensors interfaced with a microcontroller, avoiding the complexity of a full-scale BMS. Machine learning techniques are employed to analyze real-time and historical data for early prediction of abnormal behavior and battery degradation. Based on intelligent decision-making, the charging process is adapted according to battery condition, enhancing safety, efficiency, and battery lifespan. The proposed approach is well suited for educational, experimental, and small-scale EV applications, providing a practical and scalable solution for intelligent battery monitoring.

Index Terms— Electric vehicles, battery management system, predictive maintenance, machine learning, battery health monitoring, charging optimization.

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Dedicated to

Our Parents Teachers

and

Friends

Chapter-1

1.1 Introduction

The rapid growth in global energy demand and increasing concerns about environmental pollution have driven the transition toward sustainable and energy-efficient transportation systems. Electric Vehicles (EVs) have emerged as a viable alternative to conventional internal combustion engine vehicles due to their low emissions, high energy efficiency, and reduced dependence on fossil fuels. As EV adoption continues to rise worldwide, ensuring the reliability, safety, and efficiency of their battery systems has become a critical requirement for long-term vehicle performance and user satisfaction.

The battery system is one of the most important and expensive components of an Electric Vehicle. It directly affects key performance parameters such as driving range, acceleration, charging duration, and overall operational cost. Any failure or degradation in the battery system can significantly reduce vehicle performance and may lead to safety risks. Hence, continuous monitoring and accurate assessment of battery condition are essential to ensure dependable operation and to extend the service life of EV batteries.

During regular operation, EV batteries are exposed to repeated charging and discharging cycles under varying load and environmental conditions. Factors such as temperature variations, charging rate, depth of discharge, and driving patterns contribute to gradual battery degradation over time. As the battery ages, its capacity decreases and internal resistance increases, resulting in reduced efficiency and performance. Early detection of these degradation trends is necessary to prevent unexpected failures and to plan timely maintenance actions.

Conventional Battery Management Systems (BMS) generally employ fixed mathematical models and threshold-based techniques to monitor battery parameters. While these methods are relatively simple, they are often inadequate in representing the nonlinear and complex aging behavior of modern lithium-ion batteries. Such systems may provide

inaccurate estimation of battery health and fail to detect early signs of abnormal behavior, thereby limiting their effectiveness in predictive maintenance applications.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled intelligent, data-driven approaches for analyzing complex systems like EV batteries. AI and ML techniques can efficiently process large volumes of historical and real-time battery data to identify patterns associated with performance degradation and aging. These methods are particularly effective in predicting key battery parameters such as State of Health (SoH) and performance trends under different operating conditions.

The objective of this project is to develop an intelligent system that utilizes AI and ML techniques to predict the health and performance of Electric Vehicle batteries. The proposed system analyzes essential battery parameters including voltage, current, temperature, and charging behavior to assess battery condition. By learning from collected data, the model provides accurate predictions that help identify early degradation and abnormal operating behavior.

In addition to prediction, the proposed system generates alerts and warning notifications when abnormal battery behavior or excessive degradation is detected. These alerts assist the user in making informed maintenance decisions such as inspection, usage adjustment, or battery replacement. The system does not perform automatic optimization or control of battery operation; instead, it functions as a monitoring and decision-support tool aimed at improving battery safety, reliability, and longevity.

1.2 Literature Review

1. **S. S. Madani, Y. Shabeer, M. Fowler, S. Panchal, H. Chaoui, S. Mekhilef, S. X. Dou, and K. W. See, “Artificial Intelligence and Digital Twin Technologies for Intelligent LithiumIon Battery Management Syst0ems: A Comprehensive Review of State Estimation, Lifecycle Optimization, and Cloud-Edge Integration,” IEEE Access, pp. 1–20, 2024.**

Artificial Intelligence and Digital Twin Technologies for Intelligent Lithium-Ion Battery Management Systems: A Comprehensive Review of State Estimation, Lifecycle Optimization, and Cloud-Edge Integration (IEEE 1725, IEEE 2030.1)

The authors presented a comprehensive review of Artificial Intelligence and digital twin-based battery management systems for lithium-ion batteries. Their work focused on AI-driven state estimation, lifecycle optimization, and intelligent analysis of battery behaviour under varying operating conditions. The study highlighted the limitations of traditional threshold-based battery monitoring techniques.

From this study, we abstracted the idea that AI/ML techniques are more effective in predicting battery health and degradation trends. We adopted the concept of data-driven battery health prediction using parameters such as voltage, current, and temperature. This forms the foundation for implementing predictive maintenance in our project.

2. **A.F. Omojola, C. O. Ilabija, C. I. Onyeka, J. I. Ishiwu, T. G. Olaleye, I. J. Ozoemena, and P. U. Nzereogu, “Artificial Intelligence-Driven Strategies for Advancing Lithium-Ion Battery Performance and Safety,” IEEE Transactions on Energy Conversion, vol. 39, no. 2, pp. 1100–1112, 2024. 8**

Artificial Intelligence-Driven Strategies for Advancing Lithium-Ion Battery Performance and Safety (IEEE 1679.1, IEEE 1725)

This work focused on enhancing lithium-ion battery performance and safety using Artificial Intelligence techniques. The authors demonstrated that machine learning models can analyze electrical and thermal parameters to detect abnormal operating conditions. Their approach emphasized early fault detection to prevent battery damage and safety hazards.

From this reference, we abstracted the importance of continuous monitoring of voltage, current, and temperature for battery safety. We adopted the idea of generating early alerts when abnormal battery behavior is detected. This supports preventive maintenance in the proposed system.

3. **B. Bennehalli, L. Singh, S. Stephen, P. V. Prasad, B. Mallala, and A. P. C. Rao, “Machine Learning Approach to Battery Management and Balancing Techniques for Enhancing Electric Vehicle Battery Performance,” IEEE Transactions on Transportation Electrification, vol. 10, no. 1, pp. 305–315, 2023.**

Machine Learning Approach to Battery Management and Balancing Techniques for Enhancing Electric Vehicle Battery Performance (IEEE 2030, IEEE 1812)

The authors proposed a machine learning-based approach for battery management and cell balancing to improve electric vehicle battery performance. Their work showed that learning-based methods adapt better to changing battery conditions compared to rule-based techniques. The study emphasized intelligent decision support in battery systems.

From this study, we abstracted the concept that machine learning models can learn battery behaviour patterns from operational data. Although cell balancing is not implemented, we adopted the idea of ML-based battery condition assessment. This supports intelligent monitoring without requiring a full-scale Battery Management System.

4. **V. S. Naresh, G. V. N. S. R. R. Rao, and D. V. N. Prabhakar, “Predictive Machine Learning in Optimizing the Performance of Electric Vehicle Batteries: Techniques, Challenges, and Solutions,” IEEE Access, vol. 12, pp. 12500–12515, 2024.**

Predictive Machine Learning in Optimizing the Performance of Electric Vehicle Batteries: Techniques, Challenges, and Solutions (IEEE 1857, IEEE 2030.5)

This research investigated predictive machine learning techniques for optimizing electric vehicle battery performance. The authors focused on forecasting battery degradation trends and performance behavior using regression-based models. The study highlighted the shift from reactive maintenance to predictive maintenance strategies.

From this reference, we abstracted the concept of predictive maintenance using machine learning. We adopted the idea of predicting battery performance trends instead of reacting after failure. This improves system reliability and supports early maintenance planning.

- 5. N. R. Kolhalkar, G. Chhabra, N. K. Chopra, B. Kothapalli, P. Veeramanikandan, P. S. Ranjit, and L. Ziora, “Optimizing Battery Maintenance in Electric Vehicles Using Machine Learning,” IEEE Transactions on Intelligent Vehicles, vol. 8, no. 3, pp. 2401–2412, 2024**

Optimizing Battery Maintenance in Electric Vehicles Using Machine Learning (IEEE 3007.2, IEEE 1679.1)

The authors focused on optimizing battery maintenance strategies in electric vehicles using machine learning techniques. Their study demonstrated that predictive maintenance reduces unexpected failures and maintenance cost. The work emphasized AI-based systems as decision-support tools rather than automatic controllers.

From this study, we abstracted the idea of user-assisted predictive maintenance. We adopted the approach of providing battery health assessment and alerts while leaving maintenance actions to the user. This keeps the system simple, safe, and suitable for academic implementation.

Chapter 2

2.1 Objectives

- To monitor battery voltage, current and temperature in real time.
- To predict possible faults and degradation patterns using AI
- To observe changes in battery behavior and optimize charging.

2.2 Methodology

2.2.1 Hardware components

- Charging module (TP4056)



Fig.1

The TP4056 is a linear lithium-ion battery charging module designed for safe and efficient charging of single-cell Li-ion batteries. It provides constant-current and constant-voltage (CC–CV) charging, ensuring protection against overcharging. The module is compact, lowcost, and widely used in portable and low-power applications, making it suitable for small-scale EV battery monitoring systems and educational prototypes.

- Temperature sensor (DS18B20)



Fig.2

The DS18B20 is a digital temperature sensor that offers high accuracy and reliable temperature measurement using a one-wire communication protocol. It eliminates the need for external ADCs and supports multiple sensors on a single data line. In the proposed system, the DS18B20 continuously monitors battery temperature to detect abnormal thermal conditions and ensure safe battery operation.

- Voltage sensor

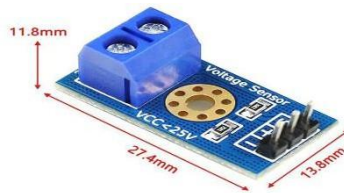


Fig:3

The voltage sensor is used to measure the battery terminal voltage in real time. Accurate voltage monitoring is essential for assessing the state of charge (SoC) and identifying overvoltage or undervoltage conditions. The measured voltage data supports battery health analysis and enables timely alerts for preventive maintenance.

- current sensor (INA219)

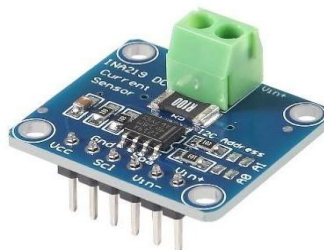


Fig:4

The INA219 is a high-precision current and power monitoring sensor that communicates with the microcontroller through the I²C interface. It measures both current flow and bus voltage, allowing real-time monitoring of battery charging and discharging behavior. The sensor provides reliable data required for battery performance analysis and fault detection.

- Microcontroller (ESP32)



Fig:5

The ESP32 serves as the core processing unit of the system. It integrates a dual-core processor with built-in Wi-Fi capability, enabling real-time data acquisition and wireless communication. The ESP32 collects sensor data, formats it into structured packets, and transmits it to a server for AI/ML-based analysis, ensuring scalability and connectivity.

- Li-ion battery cell



Fig:6

The lithium-ion battery cell acts as the energy storage element of the system. Due to its high energy density and long cycle life, it is widely used in electric vehicles and portable electronics. Monitoring key parameters of the Li-ion cell is essential to maintain safety, performance, and longevity.

- Power supply

The power supply provides regulated voltage to the ESP32, sensors, and charging module. A stable and reliable power source is crucial to ensure accurate sensor readings and uninterrupted system operation. It enables continuous monitoring and data transmission without system instability

2.2.2 Software Framework and Workflow

- Data Acquisition and Firmware Control

The ESP32 microcontroller is programmed using embedded C/Arduino framework to interface with voltage, current, and temperature sensors. The firmware continuously reads real-time sensor values at predefined intervals. Basic preprocessing such as noise filtering and range checking is performed at the device level to ensure reliable data.

- Data Transmission to Cloud

The processed sensor data is transmitted wirelessly using Wi-Fi communication. HTTP/REST APIs are used to send the data securely to the cloud server. The ESP32 acts as a client and periodically pushes sensor readings to the backend application.

- Backend Processing using Flask

A Flask-based Python server is used as the backend framework. It receives incoming data from the ESP32 and stores it in a structured format. Flask handles request routing, data validation, and communication between the hardware layer and the AI/ML model.

- AI/ML Model Integration

The received data is fed into a trained machine learning model. The model analyzes patterns in voltage, current, and temperature to detect abnormal

behavior, predict degradation, and identify potential faults. This predictive analysis enables early detection of unsafe or inefficient operating conditions.

- **Decision Logic and Optimization**

Based on the ML model output, the system determines the health status of the battery/system. If abnormal conditions are detected, control decisions such as charging optimization or alert triggering are executed. This improves safety, efficiency, and system lifespan.

- **Visualization and Alert Generation**

The processed results are displayed through a user interface or dashboard. Alerts and status notifications are generated when predefined thresholds are exceeded. This allows users to monitor system health in real time and take preventive action when required.

Block Diagram

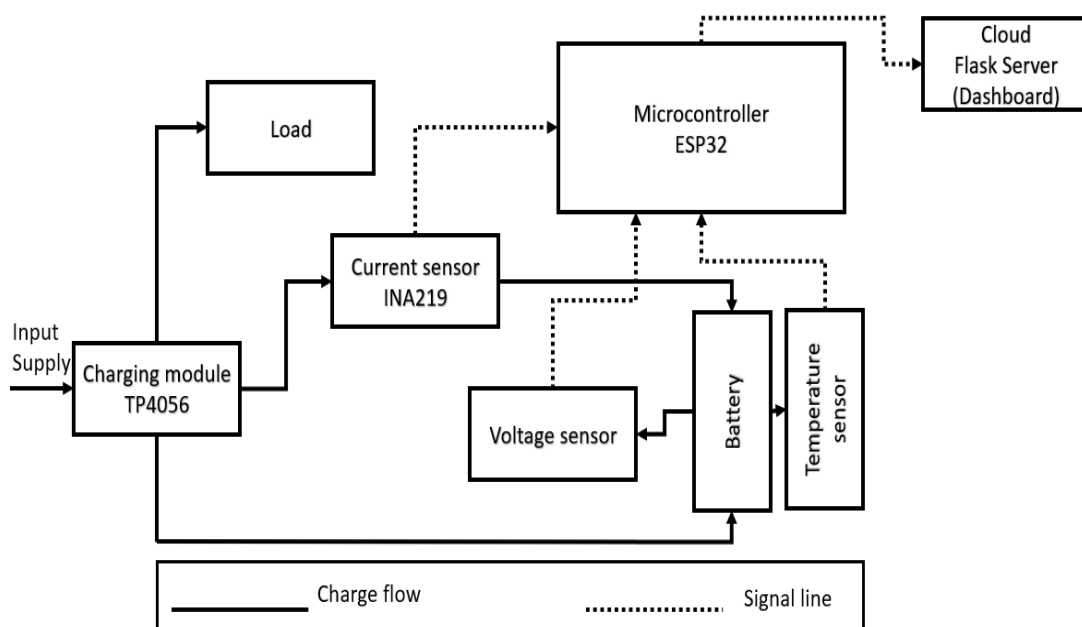


Fig 7: Block Diagram

Methodology

The proposed system continuously monitors key EV battery parameters such as voltage, current, and temperature using appropriate sensors interfaced with a microcontroller. The collected real-time data is processed and analyzed using Artificial Intelligence and Machine Learning techniques to learn normal battery operating behavior. By comparing real-time values with learned patterns, the system is capable of identifying abnormal conditions that may indicate early signs of battery stress or degradation.

Based on this intelligent analysis, the system predicts potential battery faults or health deterioration in advance and generates alert or warning messages for the user. These alerts enable users to take timely preventive maintenance actions, thereby improving battery safety and reliability. Early fault prediction helps in reducing unexpected battery failures and minimizes the risk of damage to both the battery and the vehicle.

In addition, the system maintains a continuous record of battery operating data, which can be useful for analyzing long-term battery performance trends and degradation patterns. This historical data supports better understanding of usage behavior and battery aging characteristics over time. The proposed system is designed strictly for monitoring and fault prediction and does not perform any charging control or optimization, leaving all maintenance and corrective decisions entirely to the user.

Chapter-3

3.1 Results

An AI/ML-based system has been successfully developed to continuously monitor and analyze key Electric Vehicle (EV) battery parameters such as voltage, current, and temperature in real time. Sensor data collected from the battery is processed through a microcontroller-based platform to ensure accurate and continuous observation of battery operating conditions. This real-time monitoring helps capture variations in battery behaviour under different load and environmental conditions, forming a reliable dataset for further analysis.

Using Machine Learning techniques, the system learns the normal operating behaviour of the EV battery and establishes patterns that represent healthy battery performance. By comparing real-time data with these learned patterns, the system is able to predict potential battery faults, abnormal operating conditions, and early signs of health degradation. This predictive capability allows issues such as overheating, abnormal voltage levels, or unusual current flow to be detected at an early stage, reducing the risk of sudden battery failure.

Whenever abnormal or unsafe conditions are identified, the system generates timely alert or warning notifications and communicates them to the user. These alerts enable users to take appropriate preventive maintenance actions in advance, such as inspection or servicing, thereby improving battery safety, reliability, and overall lifespan. The system focuses solely on prediction and user awareness and does not directly control or optimize the charging process, making it simple, cost-effective, and well suited for educational, experimental, and small-scale EV applications.

Main Model:

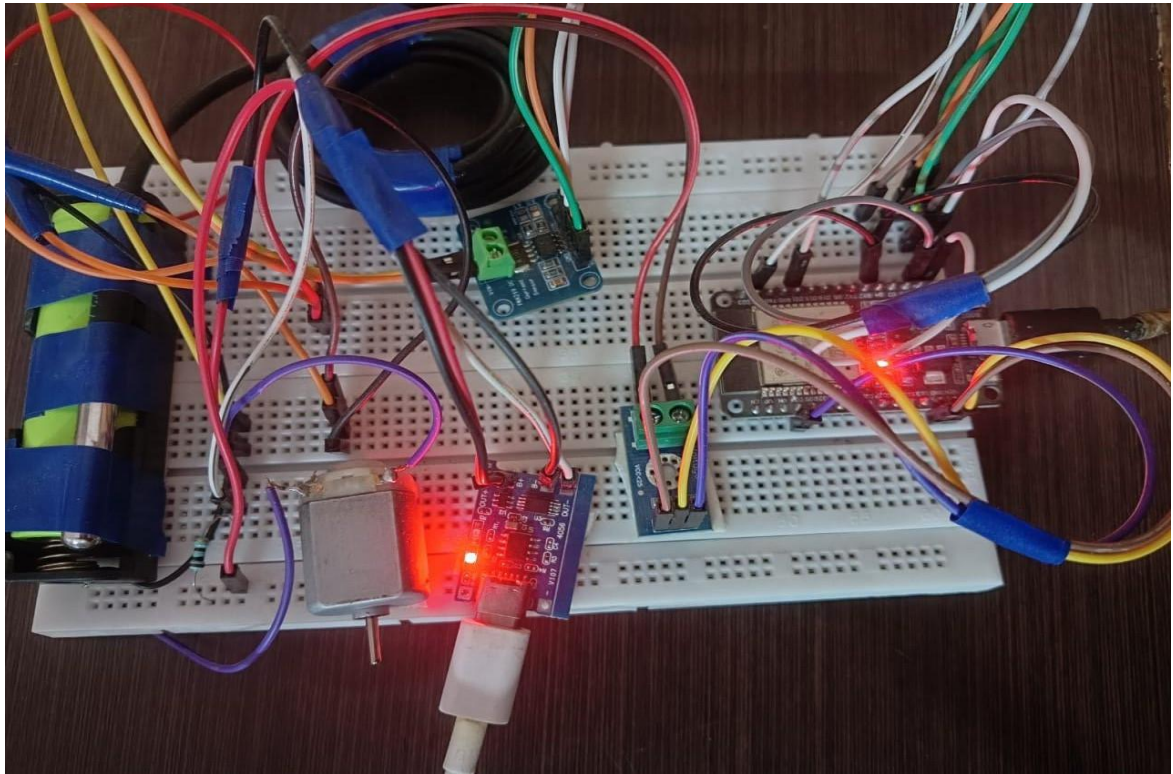


Fig3.1.1: Main Model

Dashboard block:

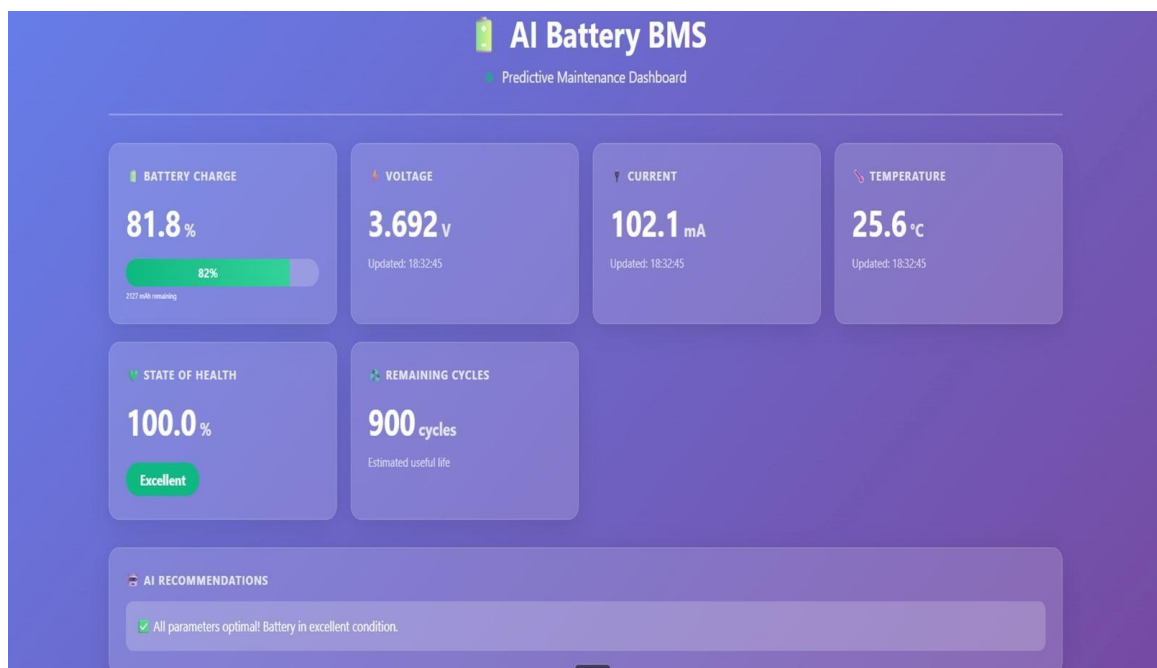


Fig3.1.2: Dashboard

3.2 Discussions:

The proposed project effectively demonstrates the application of Artificial Intelligence and Machine Learning techniques for real-time monitoring and predictive analysis of EV battery health. By continuously measuring key parameters such as voltage, current, and temperature, the system provides a detailed understanding of battery operating conditions under different scenarios. Unlike conventional threshold-based systems, the AI/ML-driven approach is capable of learning normal battery behavior and identifying subtle deviations, enabling early detection of potential faults and degradation.

The Machine Learning model processes real-time and historical sensor data to predict abnormal or unsafe operating conditions and generates timely alerts for the user. These alerts support preventive maintenance actions, helping to avoid sudden battery failures and improve overall system reliability. The system's design is simple and cost-effective, making it suitable for small-scale EV applications, laboratory experiments, and academic projects where complex Battery Management Systems may not be practical.

Overall, the project bridges the gap between theoretical AI/ML concepts and practical EV battery monitoring applications. While the current implementation focuses on prediction and alert generation, the discussion highlights the potential for future enhancements such as automatic optimization, intelligent charging control, and cloud-based monitoring. The successful results confirm that AI/ML-based battery monitoring is a promising and scalable approach for improving battery safety, performance, and lifespan.

3.3 Conclusion:

The project successfully demonstrates the design and implementation of an AI/ML-based EV battery monitoring system capable of continuously tracking critical parameters such as voltage, current, and temperature in real time. By learning normal battery behavior through Machine Learning techniques, the system effectively predicts early signs of faults and health degradation, enabling timely alert and warning notifications to the user.

The results confirm that the proposed approach improves battery safety, reliability, and lifespan by supporting preventive maintenance rather than relying solely on fixed threshold-based methods. The system's simple, cost-effective, and scalable design makes it well suited for academic, experimental, and small-scale EV applications.

Overall, the project validates the practical feasibility of integrating AI/ML with sensor-based battery monitoring. It also establishes a strong foundation for future enhancements such as automatic optimization, intelligent charging control, and cloud-based monitoring, paving the way toward more advanced and intelligent Battery Management solutions.

Reference:

1. S. S. Madani, Y. Shabeer, M. Fowler, S. Panchal, H. Chaoui, S. Mekhilef, S. X. Dou, and K. W. See, "Artificial Intelligence and Digital Twin Technologies for Intelligent Lithium-Ion Battery Management Systems: A Comprehensive Review of State Estimation, Lifecycle Optimization, and Cloud-Edge Integration," *IEEE Access*, pp. 1–20, 2024.
2. A. F. Omojola, C. O. Ilabija, C. I. Onyeka, J. I. Ishiwu, T. G. Olaleye, I. J. Ozoemena, and P. U. Nzereogu, "Artificial Intelligence-Driven Strategies for Advancing Lithium-Ion Battery Performance and Safety," *IEEE Transactions on Energy Conversion*, vol. 39, no. 2, pp. 1100–1112, 2024.
3. B. Bennehalli, L. Singh, S. Stephen, P. V. Prasad, B. Mallala, and A. P. C. Rao, "Machine Learning Approach to Battery Management and Balancing Techniques for Enhancing Electric Vehicle Battery Performance," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, pp. 305–315, 2023.
4. V. S. Naresh, G. V. N. S. R. Rao, and D. V. N. Prabhakar, "Predictive Machine Learning in Optimizing the Performance of Electric Vehicle Batteries: Techniques, Challenges, and Solutions," *IEEE Access*, vol. 12, pp. 12500–12515, 2024.
5. N. R. Kolhalkar, G. Chhabra, N. K. Chopra, B. Kothapalli, P. Veeramanikandan, P. S. Ranjit, and L. Ziora, "Optimizing Battery Maintenance in Electric Vehicles Using Machine Learning," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 3, pp. 2401–2412, 2024.